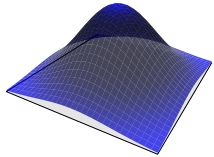
TECHNISCHE UNIVERSITÄT WIEN



Lecture 5

Statistics and Probability Theory

Covariance and Correlation
and the Law of Large Numbers and the Central Limit Theorem

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Covariance

a measure of joint variance

Covariance

The *covariance*[1] of *two real valued random variables* X and Y is

$$\mathbb{C}\mathrm{ov}(X, Y) = \mathbb{E}[(X - \mathbb{E}\,X)(Y - \mathbb{E}\,Y)]$$

the *variance*[1] is a special case

$$\mathbb{V}\mathrm{ar}(X) = \mathbb{C}\mathrm{ov}(X, X)$$

Properties: For real valued random variables X, Y, Z and constants $a, b \in \mathbb{R}$ holds

$$\mathbb{C}\mathrm{ov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y] \qquad \mathbb{V}\mathrm{ar}(X) = \mathbb{E}[X^2] - (\mathbb{E}\,X)^2$$

$$\mathbb{C}\mathrm{ov}(X, Y) = \mathbb{C}\mathrm{ov}(Y, X) \qquad \mathbb{V}\mathrm{ar}(aX) = a^2\,\mathbb{V}\mathrm{ar}(X)$$

$$\mathbb{C}\mathrm{ov}(aX + b, Y) = a\,\mathbb{C}\mathrm{ov}(X, Y) \qquad \mathbb{V}\mathrm{ar}(X + Y)$$

$$\mathbb{C}\mathrm{ov}(X + Y, Z) = \mathbb{C}\mathrm{ov}(X, Z) + \mathbb{C}\mathrm{ov}(Y, Z) \qquad = \mathbb{V}\mathrm{ar}(X) + \mathbb{V}\mathrm{ar}(Y) + 2\,\mathbb{C}\mathrm{ov}(X, Y)$$

[1] The sample versions in `R` are `cov` and `var`. Note: unbiased estimators with scaling $n - 1$.

Covariance

If X and Y are *independent*, then

$$\mathbb{C}\mathrm{ov}(X, Y) = 0$$

For independent X, Y holds $\mathbb{E}[XY] = \mathbb{E}[X]\,\mathbb{E}[Y]$ which gives

$$\mathbb{C}\mathrm{ov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y] = \mathbb{E}[X]\,\mathbb{E}[Y] - \mathbb{E}[X]\,\mathbb{E}[Y] = 0$$

The *reverse* is generally *wrong!*

For example consider $X \sim \mathrm{U}(-1, 1)$ and $Y = X^2$. Since Y is a (non-trivial) function of X the two r.v. are *not* independent, but

$$\mathbb{C}\mathrm{ov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y] = \mathbb{E}[X^3] - 0 \cdot \mathbb{E}[Y] = \frac{1}{2}\int_{-1}^{1} x^3 \,\mathrm{d}x = 0$$

The covariance $\mathbb{C}\mathrm{ov}(X, Y)$ measures the *linear dependence between X and Y.*

Correlation Coefficient

"normalized" covariance

A measure of *linear dependence* between random variables.

Correlation Coefficient

The *correlation coefficient*[1] between *two real valued random variables* is defined as

$$\rho(X, Y) = \frac{\mathbb{C}\text{ov}(X, Y)}{\sqrt{\mathbb{V}\text{ar}(X)}\sqrt{\mathbb{V}\text{ar}(Y)}}$$

Properties: It is the covariance of X, Y scaled to variance 1 or standardized;

$$\rho(X, Y) = \mathbb{C}\text{ov}\left(\frac{X}{\sqrt{\mathbb{V}\text{ar}(X)}}, \frac{Y}{\sqrt{\mathbb{V}\text{ar}(Y)}}\right) = \mathbb{C}\text{ov}\left(\frac{X - \mathbb{E}\,X}{\sqrt{\mathbb{V}\text{ar}(X)}}, \frac{Y - \mathbb{E}\,Y}{\sqrt{\mathbb{V}\text{ar}(Y)}}\right)$$

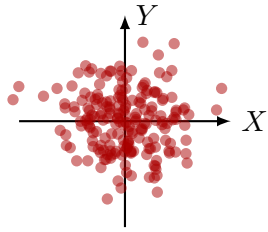
Its bounded $-1 \leq \rho(X, Y) \leq 1$, and it holds

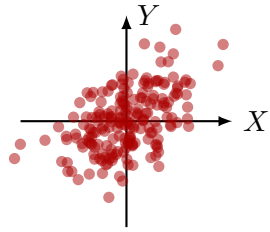
$$\rho(X, Y) = \pm 1 \qquad \Longleftrightarrow \qquad Y = aX + b \text{ with } \operatorname{sign}(a) = \rho(X, Y)$$

[1] This is the *Pearson correlation coefficient*, there are others. `R`'s sample version is `cor`.

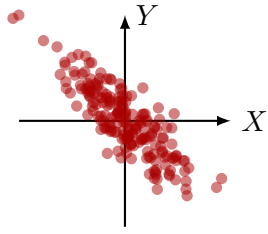
Correlation Coefficient

$\rho(X,Y) = 0$

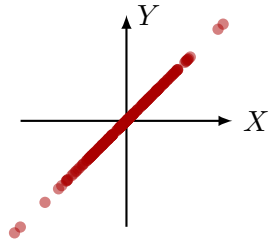




$\rho(X,Y) = 0.5$



$\rho(X,Y) = -0.8$



$\rho(X,Y) = 1$

Covariance and Correlation

We toss a fair coin three times. Let X *be the number of heads in the first two* throws and Y *be the number of heads in the last two*.

What is $\mathbb{C}\mathrm{ov}(X,Y)$ and $\rho(X,Y)$?

$$\Omega = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

with joint probability table

$X \setminus Y$	0	1	2	$P(X=x)$
0	$\frac{1}{8}$	$\frac{1}{8}$	0	$\frac{1}{4}$
1	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{1}{8}$	$\frac{1}{2}$
2	0	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{4}$
$P(Y=y)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$	1

From the marginal distributions we get $\mathbb{E}\,X = \mathbb{E}\,Y = 1$. With

$$\mathbb{E}[XY] = \sum_{x=0}^{2}\sum_{y=0}^{2} xyP(X=x, Y=y) = \frac{5}{4}$$

we get

$$\mathbb{C}\mathrm{ov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y] = \frac{1}{4}.$$

Using $\mathbb{V}\mathrm{ar}(X) = \mathbb{V}\mathrm{ar}(Y) = \frac{1}{2}$ gives

$$\rho(X,Y) = \frac{\mathbb{C}\mathrm{ov}(X,Y)}{\sqrt{\mathbb{V}\mathrm{ar}(X)}\sqrt{\mathbb{V}\mathrm{ar}(Y)}} = \frac{1}{2}$$

Covariance and Correlation

For completenes;

The pmf of the two dimensional random variable (X, Y), i.e. the values in the joint probability table are obtained as follows;

$$P(X=0,Y=0) = P(\{TTT\}) = 1/8$$
$$P(X=0,Y=1) = P(\{TTH\}) = 1/8$$
$$P(X=0,Y=2) = 0$$
$$P(X=1,Y=0) = 0$$
$$P(X=1,Y=1) = P(\{HTH, THT\}) = 2/8$$
$$P(X=1,Y=2) = P(\{THH\}) = 1/8$$
$$P(X=2,Y=0) = P(\{HTT\}) = 1/8$$
$$P(X=2,Y=1) = P(\{HHT\}) = 1/8$$
$$P(X=2,Y=2) = P(\{HHH\}) = 1/8$$

Expectation and variance computations

$$\mathbb{E}[X] = \sum_{x=0}^{2} xP(X=x) = 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{4} = 1$$

$$\mathbb{E}[X^2] = \sum_{x=0}^{2} x^2 P(X=x) = 1 \cdot \frac{1}{2} + 4 \cdot \frac{1}{4} = \frac{3}{2}$$

$$\mathbb{V}\mathrm{ar}(X) = \mathbb{E}[X^2] - (\mathbb{E}\,X)^2 = \frac{3}{2} - 1 = \frac{1}{2}$$

Covariance and Correlation

We toss a fair coin three times. Let X *be the number of heads in the first two* throws and Y *be the number of heads in the last two*.

What is $\mathbb{C}\mathrm{ov}(X,Y)$ and $\rho(X,Y)$? Alternative Approach:
Let $B_i \sim \text{Bernoulli}(\frac{1}{2})$ be the outcome of the ith toss for $i \in \{1,2,3\}$.
The individual B_i's are independent and $X = B_1 + B_2$ and $Y = B_2 + B_3$, then

$$\begin{aligned}\mathbb{C}\mathrm{ov}(X,Y) &= \mathbb{C}\mathrm{ov}(B_1+B_2, B_2+B_3)\\ &= \mathbb{C}\mathrm{ov}(B_1,B_2) + \mathbb{C}\mathrm{ov}(B_1,B_3) + \mathbb{C}\mathrm{ov}(B_2,B_2) + \mathbb{C}\mathrm{ov}(B_2,B_3)\\ &= \mathbb{C}\mathrm{ov}(B_2,B_2) = \mathbb{V}\mathrm{ar}(B_2) = \frac{1}{4}\end{aligned}$$

Since $\mathbb{V}\mathrm{ar}(X) = \mathbb{V}\mathrm{ar}(B_1+B_2) \overset{\text{ind}}{=} \mathbb{V}\mathrm{ar}(B_1) + \mathbb{V}\mathrm{ar}(B_2) = \frac{1}{2}$ we get

$$\rho(X,Y) = \frac{\mathbb{C}\mathrm{ov}(X,Y)}{\sqrt{\mathbb{V}\mathrm{ar}(X)}\sqrt{\mathbb{V}\mathrm{ar}(Y)}} = \frac{1/4}{\sqrt{1/2}\sqrt{1/2}} = \frac{1}{2}.$$

Random Samples

and the Sample Mean

Let $X_1, X_2, \ldots, X_n$ be n *independent and identically distributed* (iid) random variables. This is known as a *random sample*.

Sample Mean

For a *random sample* $X_1, \ldots, X_n$, the *sample mean* is defind as

$$\overline{X}_n = \frac{1}{n} \sum_{i=1}^{n} X_i$$

Properties: If $\mathbb{E}\, X_1 = \mu$ and $\mathbb{V}\mathrm{ar}(X_1) = \sigma^2$, then

$$\mathbb{E}\, \overline{X}_n = \mu \qquad \text{and} \qquad \mathbb{V}\mathrm{ar}(\overline{X}_n) = \frac{\sigma^2}{n}$$

If $X_i \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ for $i = 1, \ldots, n$ then $\overline{X}_n \sim \mathcal{N}(\mu, \frac{\sigma^2}{n})$.
In words, the *sample mean* of a *normal distributed random sample* is also *normal*.

Law of Large Numbers

The *Law of Large Numbers* stats that the sample mean $\overline{X}_n$ will be *with high probability* very *close to the expectation* μ of the underlying distribution.

Law of Large Numbers

Let $X_1, X_2, \ldots$ *be a sequence of* iid *random variables* with expectation $\mathbb{E}\, X_1 = \mu$ and *finite variance* $\mathbb{V}\mathrm{ar}(X_1) = \sigma^2 < \infty$. Denote with $\overline{X}_n$ the *sample mean of the first* n random variables, then

$$\lim_{n\to\infty} P(|\overline{X}_n - \mu| < \epsilon) = 1 \qquad \text{for all } \epsilon > 0.$$

In other words, the sample mean of an iid random sample with finite variance *converges in probability* to the mean $\overline{X}_n \xrightarrow{p} \mathbb{E}\, X_1$.

Law of Large Numbers

Consider a *discrete random variable* X with pmf $f_X(x) = P(X = x)$ given by

x	1	2	3	4
$f_X(x)$	0.1	0.3	0.2	0.4

Its expectation and variance is given by

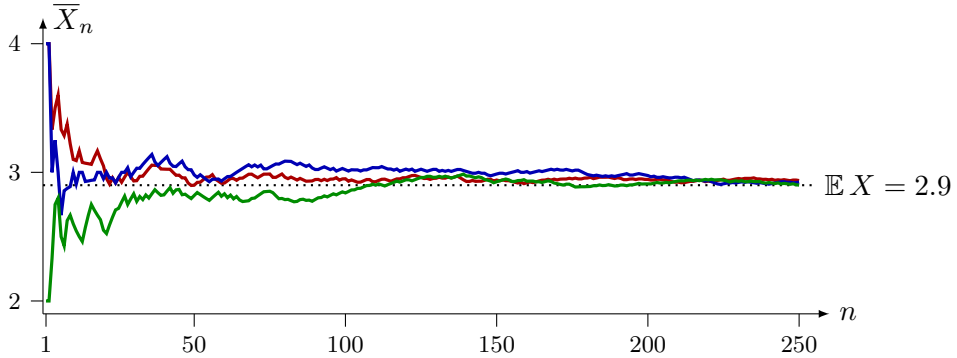
$$\mathbb{E}\,X = \sum_{x=1}^{4} x f_X(x) = 2.9, \qquad \mathbb{V}\mathrm{ar}(X) = \sum_{x=1}^{4} (x - \mathbb{E}\,X)^2 f_X(x) = 1.09 < \infty.$$

For a sequence of random variables $X_i \overset{\text{iid}}{\sim} f_X$, the Law of Large Numbers is applicable

$$\overline{X}_n = \frac{1}{n} \sum_{i=1}^{n} X_i \xrightarrow{p} \mathbb{E}\,X = 2.9$$

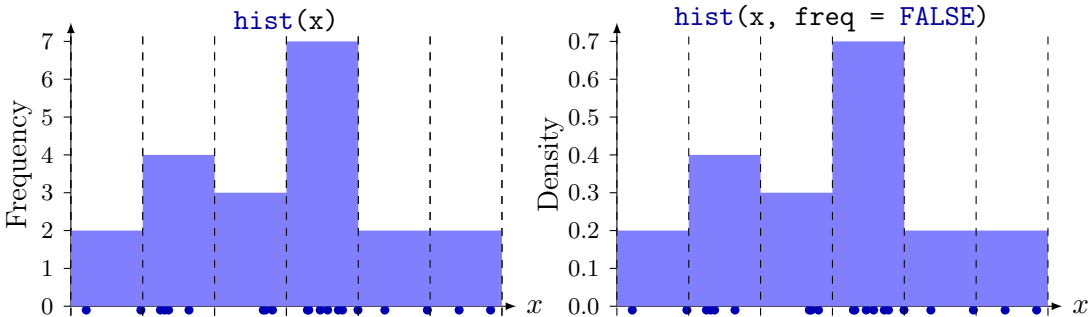
Consider a *discrete random variable* X with pmf $f_X(x) = P(X = x)$ given by

x	1	2	3	4
$f_X(x)$	0.1	0.3	0.2	0.4



Histograms

A *histogram* is a visual representation of the distribution of *observations* x_i, *organized into bins or intervals* where the *height of each bin corresponds to the count* of observations within that interval. There are two versions

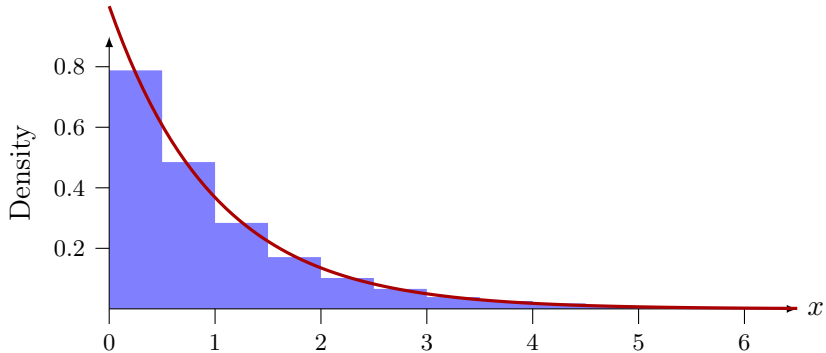


FREQUENCY: The bar height is the number of observations within the bins.

DENSITY: The total area under the bars in 1, allows comparison to pmf or density.

Histograms

The *Law of Large Numbers* implies that the *density histogram*[2] approaches the underlying *density*.



Density histogram with bin width $\frac{1}{2}$ of 10 000 random draws from Exp(1) and *density* of Exp(1).

[2] In R, use `hist(x, freq = FALSE)`

Central Limit Theorem

Central Limit Theorem[3]

Let $X_1, X_2, \ldots$ *be a sequence of iid random variables* with *finite variance* $\mathbb{V}\mathrm{ar}(X_1) < \infty$ and $\overline{X}_n$ be the *sample mean of the first n random variables*, then

$$\sqrt{n}(\overline{X}_n - \mathbb{E}\,X_1) \xrightarrow{d} \mathcal{N}(0, \mathbb{V}\mathrm{ar}(X_1))$$

In practice, for *large n*, the *sample mean is approximately normal*

$$\overline{X}_n \approx \mathcal{N}(\mu, \sigma^2/n)$$

Similarly, for the sum $S_n = X_1 + X_2 + \ldots + X_n = n\overline{X}_n$ we have

$$S_n \approx \mathcal{N}(n\mu, n\sigma^2)$$

Thus, the *standardized $\overline{X}_n$ and S_n* are *approximately standard normal*

$$\frac{\overline{X}_n - \mu}{\sigma/\sqrt{n}} \approx \mathcal{N}(0,1) \qquad \text{and} \qquad \frac{S_n - n\mu}{\sqrt{n}\sigma} \approx \mathcal{N}(0,1).$$

[3] In short CLT

CLT Simulation

Let $X_1, X_2, \ldots$ be iid Exp(1) distributed.

$$\mathbb{E}\, X_1 = \mu = 1 \qquad \text{and} \qquad \mathbb{V}\text{ar}(X_1) = \sigma^2 = 1 < \infty.$$

By the CLT, the standardized sample mean $\overline{X}_n$ is (for large n) approximately standard normal

$$\sqrt{n}(\overline{X}_n - 1) = \frac{\overline{X}_n - \mu}{\sigma/\sqrt{n}} \approx \mathcal{N}(0, 1)$$

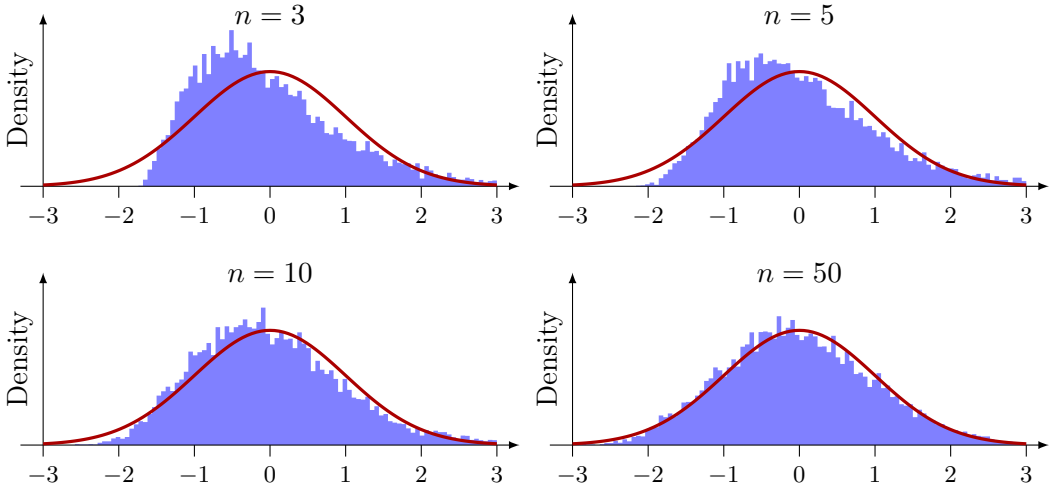
(Run R simulation)

CLT Simulation

Exponential

Histograms of standardized means $\sqrt{n}(\overline{X}_n - 1)$ with *standard normal density in red*

Based on 10 000 simulations of the mean $\overline{X}_n$



Normal Approximation of the Binomial

The sum $S_n = X_1 + \ldots + X_n$ for $X_i \overset{\text{iid}}{\sim} \text{Bernoulli}(p)$ follows a Binomial distribution

$$S_n \sim \text{Binomial}(n, p)$$

With the individual Bernoulli trials having expectation and variance

$$\mathbb{E}\, X_1 = p \qquad \text{and} \qquad \mathbb{V}\text{ar}(X_1) = p(1-p)$$

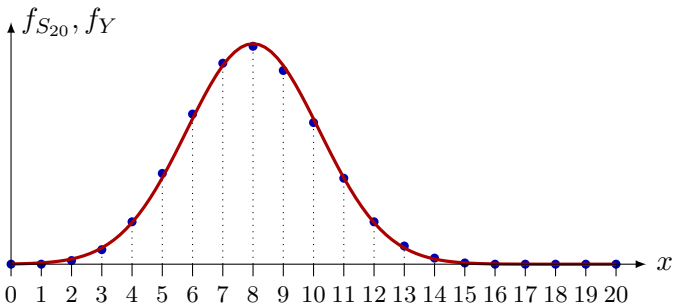
we get using the CLT that

$$S_n \approx \mathcal{N}(np, np(1-p))$$

Normal Approximation of the Binomial

$$\text{Binomial}(n,p) \approx \mathcal{N}(np, np(1-p))$$

Plotting the pmf $f_{S_{20}}$ of $S_{20} \sim \text{Binomial}(20, 0.4)$ with the normal density f_Y for $Y \sim \mathcal{N}(20 \cdot 0.4, 20 \cdot 0.4 \cdot 0.6)$



Allows to (approximately) compute probability ranges based on the normal cdf Φ

Normal Approximation of the Binomial

$\text{Binomial}(n,p) \approx \mathcal{N}(np, np(1-p))$

To compute the the probability of $a \leq S_n \leq b$ for $a, b \in \{0, 1, \ldots, n\}$ we can normalize the Binomial random variable $S_n \sim \text{Binomial}(n,p)$ to get

$$Z_n = \frac{S_n - np}{\sqrt{np(1-p)}} \approx \mathcal{N}(0,1)$$

This means that

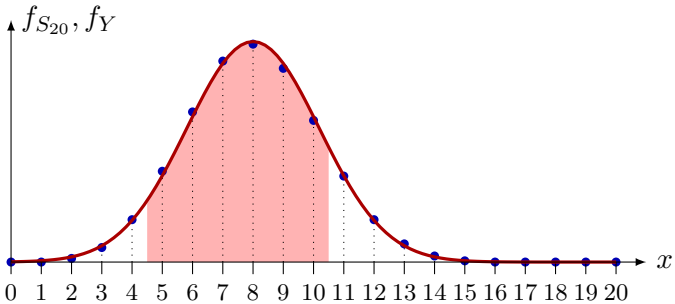
$$P(a \leq S_n \leq b) \approx \Phi\left(\frac{b + \frac{1}{2} - np}{\sqrt{np(1-p)}}\right) - \Phi\left(\frac{a - \frac{1}{2} - np}{\sqrt{np(1-p)}}\right)$$

with the addition/subtraction of $\frac{1}{2}$ is known as *continuity correction*.
Rule of thumb: Approximation is considered reasonable, if $10 \leq \min(np, n(1-p))$.

Normal Approximation of the Binomial

$\text{Binomial}(n, p) \approx \mathcal{N}(np, np(1-p))$

$$P(a \leq S_n \leq b) \approx \Phi\left(\frac{b + \frac{1}{2} - np}{\sqrt{np(1-p)}}\right) - \Phi\left(\frac{a - \frac{1}{2} - np}{\sqrt{np(1-p)}}\right)$$



Nowadays, less of a concern! Use `R` and avoid approximations if possible.

Normal Approximation of the Poisson

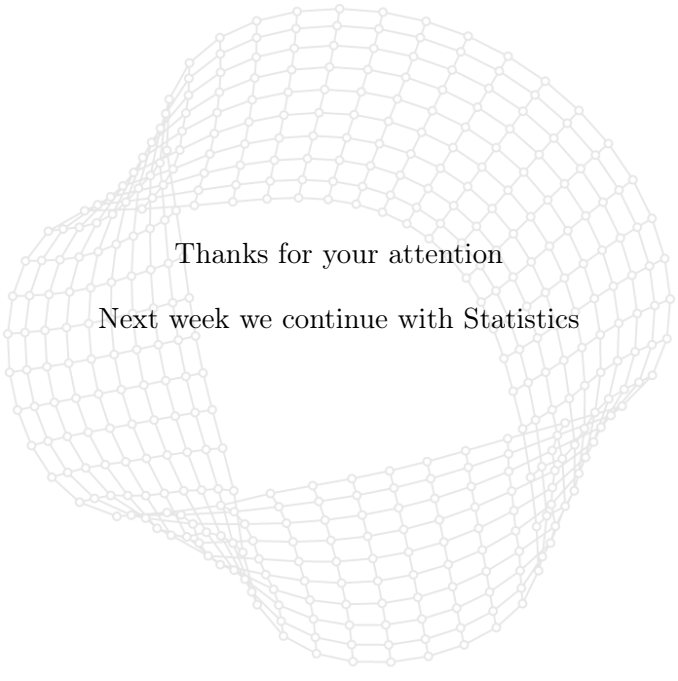
$\text{Poisson}(\lambda) \approx \mathcal{N}(\lambda, \lambda)$

Another classic Normal approximation is for the Poisson distribution.

Let $X \sim \text{Poisson}(\lambda)$ and $a, b \in \mathbb{N}_0$ such that $a \leq b$, then

$$P(a \leq X \leq b) \approx \Phi\left(\frac{b + \frac{1}{2} - \lambda}{\sqrt{\lambda}}\right) - \Phi\left(\frac{a - \frac{1}{2} - \lambda}{\sqrt{\lambda}}\right)$$

where the addition/subtraction of $\frac{1}{2}$ is the *continuity correction*.
Rule of thumb: Approximation is considered reasonable, if $\lambda > 15$.



Thanks for your attention

Next week we continue with Statistics